LEARNING MODULE

17

Law of Large Numbers and Central Limit Theorem

Averages settle and sums turn normal. Compounded wealth follows the median path, not the mean path, and TAIEX shows how slowly the laws can work.

VOLUME III · Probability

BUILDS ON LM1 · LM3 · LM4 · LM5 · LM13 · LM14 · LM15 · LM16

LEADS TO LM18 · LM19 · LM20 · LM21 · LM22 · LM23 · LM24 · LM25 · LM26 · LM34 · LM36 · LM38 · LM41 · LM42 · LM45

TIME ≈ 3 h + lab

LEARNING OUTCOMES

Mastery	*After this module you should be able to:*
☐	**1** calculate the mean and variance of sums and averages of independent random variables, and explain why the standard deviation of an average falls like $1/\sqrt{n}$
☐	**2** derive the weak law of large numbers from Chebyshev's inequality, state the conditions of the weak and strong laws, and apply them to frequencies and to the binary bet of LM4
☐	**3** explain why realized log growth converges to $g = \mathbb{E}[\ln(1+R)]$ while expected wealth compounds at $\ln(1+\mu)$, and why the chance of ending above expected wealth tends to zero
☐	**4** state the central limit theorem with its conditions, and apply the normal approximation to binomial counts
☐	**5** compute the median, quantiles and loss probability of wealth under the lognormal approximation, with and without leverage
☐	**6** measure how fast sums of TAIEX log returns approach the normal shape, and explain how dependence slows the approach
☐	**7** identify when the laws fail (no mean, infinite variance, infinite fourth moment, dependence) and what each failure does to a sample statistic
☐	**8** compute running averages, sums over windows of m days and normal approximations in Python, counting threshold days with the book's guard

LEARNING MODULE OVERVIEW

- The average of n uncorrelated draws keeps their mean and has variance σ_X^2/n, (17.1): its standard deviation falls like $1/\sqrt{n}$. Identical copies do not average out at all.
- Chebyshev's inequality turns (17.1) into the weak law of large numbers, (17.2). A fair coin's frequency misses one half by 0.1 or more with probability 0.7539, 0.05689 and 2.7×10^{-10} after 10, 100 and 1,000 tosses; Chebyshev's bound is 2.5, 0.25 and 0.025. The strong law speaks about whole paths, with probability one.
- Realized log growth settles at $g = \mathbb{E}[\ln(1 + R)]$, (17.3), the median path. Expected wealth compounds faster, at $\ln(1 + \mu)$, and the chance of ending above it tends to zero, (17.4), slowly. Under an iid model of TAIEX's total-return days that chance is 0.32 after 23.8 years, and about 0.028 by the normal approximation at the in-sample Kelly leverage of 4.14.
- TAIEX's average day, 0.06251%, compounds to 38.7 times the start over 5,849 days, while the index rose 25.2-fold. The factor 1.534 (38.66/25.20) between them is the daily Jensen gap of 0.7316 bp, accumulated.
- The central limit theorem makes a standardized sum of iid draws with finite variance approximately normal, (17.5). The chance of being behind after 100 trades of the Kelly bet of LM4 is 0.04431 exactly, 0.04464 by the normal approximation and 0.04457 with a continuity correction.
- Wealth is therefore approximately lognormal. Under the in-sample iid model of TAIEX, the normal approximation (17.6) puts the chance of a loss at 23.74% over one year and 1.19% over ten years. Of the 23 calendar years 2003–2025, 5 lost.
- Excess kurtosis of a sum of m iid days falls like $1/m$, (17.7): from 5.428 to 0.27 at 20 days. TAIEX's overlapping 20-day log-return sums show 2.29, because large moves cluster.
- Without a mean, Cauchy averages never settle; without a variance, the $\sqrt{n}$ normal limit fails; without a fourth moment, a sample kurtosis never settles. TAIEX's running excess kurtosis jumped from 4.25 to 4.94 on 7 April 2025.

1 INTRODUCTION

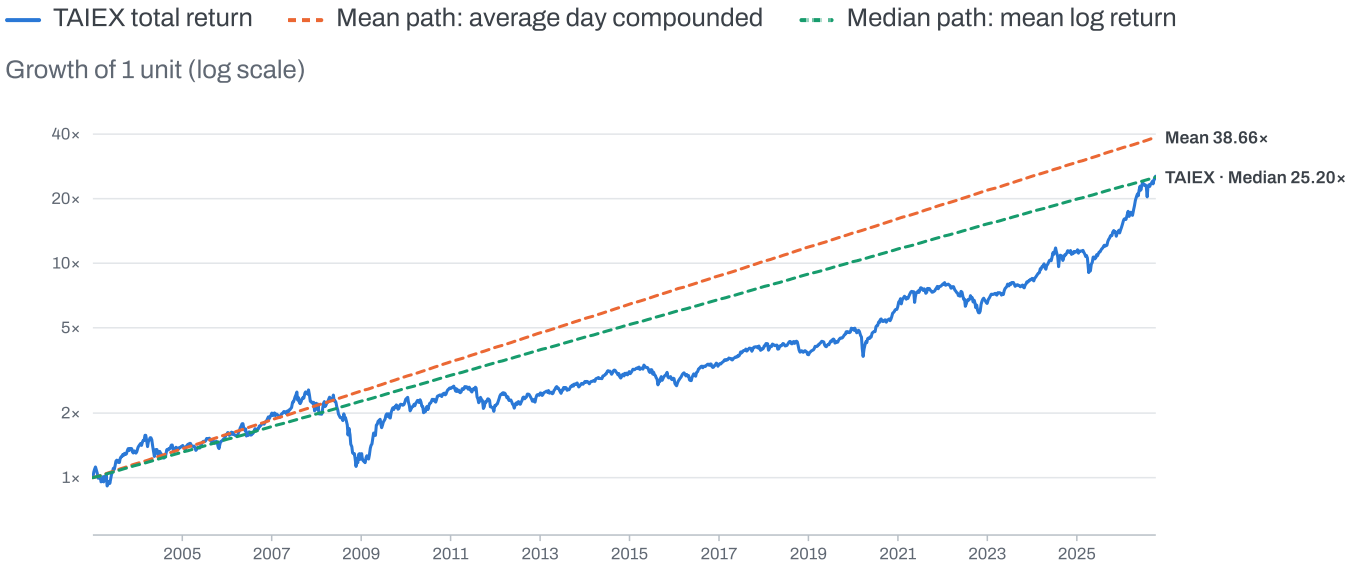
This module asks what happens to averages and sums of many independent random variables, such as a year's 246 daily returns. The law of large numbers says that an average settles at the expectation; the central limit theorem says that a standardized sum is approximately normal. The case shows what goes wrong when they are applied to the wrong quantity.

MOTIVATING CASE **Compounding the average day**

From 2 January 2003 to 8 October 2026 the TAIEX total-return index, with dividends reinvested and no costs, rose 25.2-fold over 5,849 trading days. Its average daily return was 0.06251%. Compounding that average 5,849 times gives 38.7 times the start, against the index's 25.2 (Exhibit 1).

Now apply a constant leverage f to the same days, rebalanced daily with free borrowing (a hypothetical). Compounding the average day then promises more for every f: 2.2 million times the start at $f = 4$. Realized wealth grows only up to $f = 4.14$, to 2,147 times, and is back at the start at $f = 8.04$. Which path does an investor live through, and why does the average day overstate it?

Exhibit 1: TAIEX total-return wealth against the mean path and the median path



Note: wealth P_t/P_0 of the TAIEX total-return index on a log scale, with the mean path $(1+\bar{R})^t$ and the median path $e^{t\bar{r}}$ after t daily returns. Both model paths are in-sample, using the sample's own means, so the median path ends at 25.20 by construction. Source: FinMind `TaiwanStockTotalReturnIndex` (TAIEX), retrieved 18 October 2026. Computations: `data/lm17_numbers.py` and `data/lm17_charts.py`.

Sections 2 to 4 develop the laws of large numbers and resolve the case. Sections 5 and 6 add the central limit theorem and the lognormal approximation, and Sections 7 to 9 test the laws on TAIEX.

NOTE **Notation for this module**

The data are TR23, the TAIEX total-return index: $T = 5{,}849$ daily returns (a count) from 3 January 2003 to 8 October 2026. They span 8,680 calendar days from the first close, 2 January 2003, that is 8,680/365.25 = 23.7645 years, so $A = 246.12$ sessions a year. Simple returns are R_t; log returns $r_t = \ln(1 + R_t)$ enter every sum. Moments $\bar{R}$, s, $\bar{r}$, s_r divide by T. A horizon of τ years is $m = \text{round}(A\tau)$ sessions, with wealth W_m at its end.

Generic results use μ_X, σ_X; for a daily return, μ, σ, and $g = \mathbb{E}[\ln(1 + R)]$, σ_r for its log. Local: n draws, j wins (LM4's N is n); S_n, $\bar{X}_n$ their sum and average; $Y_i = X_i - \mu_X$; Z_n the standardized sum; $\varepsilon > 0$; $\delta := \ln(1 + \mu) - g$, LM14's daily Jensen gap, which LM14 names only in words (a symbol is needed for the proof of (17.4) and for the bounds), $\delta(f)$ at leverage f; $\bar{g}(f)$ (4.4) and $s_r(f)$, mean and standard deviation of $\ln(1 + fR_t)$; $\tilde{f}$ a second position; g^* LM4's Kelly growth; $\hat{p}$ TR23's share of up days; α a tail index; $M_X(t)$ the moment generating function.

NOTE **Research hygiene**

Every TAIEX figure covers 3 January 2003 to 8 October 2026, except the calendar years 2003–2025 (2003 from its first close, 2 January 2003). TR23's path and running statistics are **descriptive**. The wealth model draws days independently from the sample's own returns: **in-sample**, a model, not a forecast. The kurtosis by limit regime is a **stability check**. Leveraged paths are **hypothetical**: daily rebalancing, free borrowing, no costs. Nothing is tested on later data (LM34).

2 SUMS AND AVERAGES OF INDEPENDENT VARIABLES

- ☐ calculate the mean and variance of sums and averages of independent random variables, and explain why the standard deviation of an average falls like $1/\sqrt{n}$

Take two independent trades of LM3's bet, each returning +2% or −1% with equal probability. By (13.6) and (13.7) the outcomes $++$, $+-$, $-+$ and $--$ each have probability $\frac{1}{4}$. The average return of the two trades is 2%, 0.5%, 0.5% or −1%. Its mean is 0.5%, as for one trade, and its variance, in %², is

$$\tfrac{1}{4}\left(1.5^2 + 0^2 + 0^2 + 1.5^2\right) = 1.125,$$

half of one trade's $\frac{1}{2}(1.5^2 + 1.5^2) = 2.25$. Averaging kept the mean and halved the variance.

Let $X_1, \ldots, X_n$ have a common mean μ_X and a common variance $\sigma_X^2 < \infty$. They are **iid**, independent and identically distributed, when they are independent ((13.13), (16.1)) and share one distribution. Their sum and average are

$$S_n := X_1 + \cdots + X_n, \qquad \bar{X}_n := \frac{S_n}{n}.$$

By (14.3), $\mathbb{E}S_n = n\mu_X$. Independent draws with finite variance are uncorrelated (LM16), so the sum rule (16.3) gives $\text{Var}\, S_n = n\sigma_X^2$, and (14.4) with $\eta = 1/n$ divides it by n^2.

RESULT **Mean and variance of an average**

Let $X_1, \dots, X_n$ be iid with mean μ_X and variance $\sigma_X^2 < \infty$, or uncorrelated with a common mean μ_X and a common variance $\sigma_X^2 < \infty$. Then

$$\mathbb{E}\bar{X}_n = \mu_X, \qquad \operatorname{Var}\bar{X}_n = \frac{\sigma_X^2}{n}. \tag{17.1}$$

The standard deviation $\sigma_X/\sqrt{n}$ of the average is its **standard error**. LM4 used it with s in place of σ_X, and LM20 derives that estimate. A hundred times as many draws cut the standard error by a factor of ten.

EXAMPLE 1

How many trades make the edge visible?

LM3's bet has mean $\mu = 0.5\%$ and standard deviation $\sigma = 1.5\%$ per trade. Find the standard error of the average after 100 and after 10,000 independent trades, and the smallest n at which it is at most 0.25%, half the edge.

SOLUTION

By (17.1), the standard error is $1.5\%/\sqrt{n}$: 0.15% after 100 trades and 0.015% after 10,000. It is at most 0.25% when $\sqrt{n} \geq 1.5/0.25 = 6$, that is, from $n = 36$.

PITFALL **Correlated days do not average out**

Formula (17.1) needs uncorrelated draws. For identical copies, $X_i = X_1$ for every i, the average is X_1 itself and keeps the variance σ_X^2. In the sum rule (16.3) the $n(n-1)$ covariances all equal σ_X^2, so $\operatorname{Var} S_n = n^2\sigma_X^2$. Positively correlated days lie in between. Whether daily variances add up over months is a question about data, which LM19 examines.

KNOWLEDGE CHECK 1

Four independent days each have a standard deviation of 1.2%. What is the standard deviation of their average return? What is it if the four days are identical copies of one day?

ANSWER

By (17.1), $1.2\%/\sqrt{4} = 0.6\%$. For identical copies the average is the single day's return, so its standard deviation stays 1.2%.

3 THE LAW OF LARGE NUMBERS

- [ ] derive the weak law of large numbers from Chebyshev's inequality, state the conditions of the weak and strong laws, and apply them to frequencies and to the binary bet of LM4

Toss a fair coin n times, with j heads. How fast does the frequency j/n approach $\frac{1}{2}$? The chance of a miss by 0.1 or more is an exact binomial sum, (15.1): 0.7539, 0.05689 and 2.7×10^{-10} at n = 10, 100 and 1,000.

Chebyshev's inequality (14.8) bounds the same chance without the pmf. By (17.1) the frequency has variance $\frac{1}{4}/n$, so the bound is $(\frac{1}{4}/n)/0.1^2 = 25/n$: 2.5, 0.25 and 0.025. The bound is crude, but it tends to zero.

DEFINITION **Convergence in probability**

Random variables $\bar{X}_n$ converge **in probability** to a number x when $\Pr(|\bar{X}_n - x| \geq \varepsilon) \to 0$ as $n \to \infty$, for every tolerance $\varepsilon > 0$.

THEOREM **Weak law of large numbers**

Let $X_1, \ldots, X_n$ be iid with mean μ_X and variance $\sigma_X^2 < \infty$, or uncorrelated with a common mean μ_X and a common variance $\sigma_X^2 < \infty$. Then for every $\varepsilon > 0$

$$\Pr\left(|\bar{X}_n - \mu_X| \geq \varepsilon\right) \leq \frac{\sigma_X^2}{n\varepsilon^2}, \tag{17.2}$$

so $\bar{X}_n$ converges in probability to μ_X.

Proof. If $\sigma_X = 0$, the average equals μ_X with probability one. Otherwise, by (17.1) it has mean μ_X and standard deviation $\sigma_X/\sqrt{n}$. A miss of ε is $\varepsilon\sqrt{n}/\sigma_X$ standard deviations, so by (14.8) its probability is at most $\sigma_X^2/(n\varepsilon^2)$. ■

Only (17.1) was used, so uncorrelated draws suffice (Durrett 2019, Theorem 2.2.3). For iid draws the weak law is Grinstead and Snell's (1997, §8.1, Theorem 8.2) and Wasserman's (2004, Theorem 5.6). For such draws a finite mean is even enough, with a longer proof (Durrett 2019, Theorem 2.2.14).

For independent trials that succeed with probability p, the indicator of the event $\mathcal{A}_i$ that trial i succeeds, $\mathbf{1}_{\mathcal{A}_i}$, has mean p by (14.1) and variance $p(1-p) \leq \frac{1}{4}$ by (14.4). The count j is their sum, so (17.2) gives

$$\Pr\left(\left|\frac{j}{n} - p\right| \geq \varepsilon\right) \leq \frac{p(1-p)}{n\varepsilon^2} \leq \frac{1}{4n\varepsilon^2}.$$

A frequency converges in probability to its probability: this is the $j/N \to p$ of LM4's binary bet.

EXAMPLE 2

How many trials pin down a frequency?

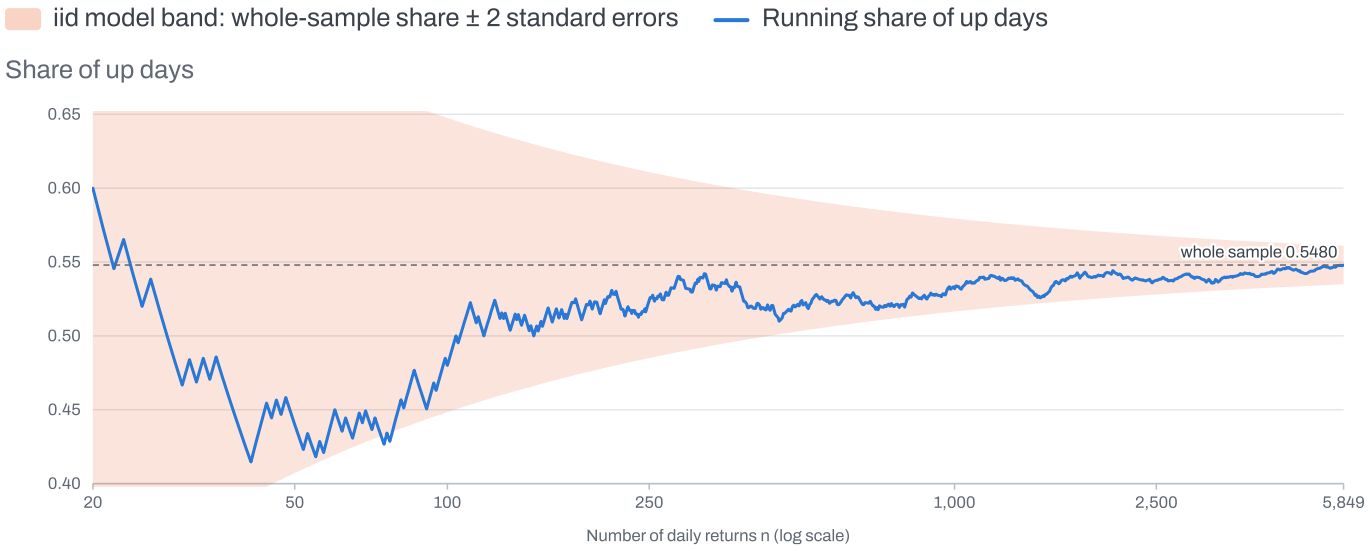
By Chebyshev's bound, how many independent trials guarantee a frequency within one percentage point of p with probability at least 95%, whatever p is?

SOLUTION

The worst case is $p = \frac{1}{2}$, where $p(1-p) = \frac{1}{4}$. The bound is at most 0.05 when $n \geq 0.25/(0.01^2 \times 0.05) = 50{,}000$. It is a property of a model of independent trials, not of trading days.

On TAIEX, 3,205 of TR23's 5,849 days rose and 2,644 fell (LM13), and exact arithmetic on the closes finds no unchanged day. Exhibit 2 plots the running share of up days: 0.5163 after 246 days, 0.5378 after 1,231 and $\hat{p} = 0.5480$ at the end.

Exhibit 2: Running share of TAIEX up days



Note: share of up days ($R_t > 10^{-12}$) among TR23's first n daily returns, logarithmic n axis. The band $\hat{p} \pm 2\sqrt{\hat{p}(1-\hat{p})/n}$, of half-width 0.0130 at $n = 5{,}849$, belongs to an iid model with p fitted to the whole sample: in-sample, and no exits are counted. Source: FinMind `TaiwanStockTotalReturnIndex` (TAIEX), retrieved 18 October 2026.

The weak law speaks about each n separately, so the weak law alone does not exclude a frequency that wanders away from p again and again. The strong law speaks about a whole path.

THEOREM **Strong law of large numbers**

Let $X_1, X_2, \ldots$ be iid with $\mathbb{E}|X_1| < \infty$ and $\mu_X = \mathbb{E}X_1$. Then $\bar{X}_n \to \mu_X$ with probability one: the paths along which the averages converge to μ_X have probability one. Stated without proof (Wasserman 2004, Theorem 5.18; proved in Durrett 2019, Theorem 2.4.1).

Convergence **with probability one**, or **almost surely**, is stronger than convergence in probability, and the strong law needs only a finite mean.

> **DEEP DIVE** **What "with probability one" adds** optional · click to expand
>
> Toss a fair coin forever. The all-heads path exists, yet its probability is at most 2^{-n} for every n, hence zero, an event of the kind LM13 described. The strong law says that all paths along which j/n fails to converge to $\frac{1}{2}$ have probability zero together. The weak law, about one n at a time, cannot say this. Grinstead and Snell (1997, §8.1, Exercise 16) give a sequence that obeys the weak law but not the strong.

LM3's Deep Dive made a weak-law argument: the sum of squared increments has mean τ and variance $2\tau\Delta t \to 0$, so by (14.8) it converges in probability to τ. LM18 simulates random paths; here every probability is exact or an analytic approximation.

> **PITFALL** **The law does not correct runs**
>
> After ten up days, independent days make a down day no more likely; believing otherwise is the **gambler's fallacy** (Blitzstein and Hwang 2019, §10.2). A frequency converges because early surpluses are diluted, not repaid: the surplus of heads $j - n/2$ even has a growing standard deviation, $\sqrt{n}/2$.

> **導讀** **大數法則：偏差被稀釋，不會被扳回**
>
> 大數法則（law of large numbers）說的是：獨立（independent）抽樣的樣本平均數（sample mean）會靠近母體（population）的期望值（expectation）。弱大數法則（weak law of large numbers）由謝比雪夫不等式（Chebyshev's inequality）證明：平均數的變異數（variance）是 σ_X^2/n，偏離超過 ε 的機率不超過 $\sigma_X^2/(n\varepsilon^2)$，這叫依機率收斂（convergence in probability）。平均數的標準差（standard deviation）$\sigma_X/\sqrt{n}$ 稱為標準誤（standard error），樣本加大一百倍，標準誤只縮小為十分之一。強大數法則（strong law of large numbers）講的是整條路徑：平均數不收斂的路徑，機率合起來是零，這叫幾乎必然收斂（almost sure convergence）。頻率會收斂，是因為早期的偏差被稀釋，不是因為之後會補回來；以為連漲之後比較容易跌，就是賭者謬誤（gambler's fallacy）。

KNOWLEDGE CHECK 2

For a fair coin, compute the standard deviations of the number and of the frequency of heads after 100 and after 10,000 tosses. Which one tends to zero?

ANSWER

The count has $\sqrt{n}/2$: 5 and 50. The frequency has $0.5/\sqrt{n}$: 0.05 and 0.005, which tends to zero.

4 GROWTH: THE MEAN PATH AND THE MEDIAN PATH

- [ ] explain why realized log growth converges to $g = \mathbb{E}[\ln(1+R)]$ while expected wealth compounds at $\ln(1+\mu)$, and why the chance of ending above expected wealth tends to zero

LM3 warned that +100% and −50% leave wealth where it started. Play that bet ten times.

EXAMPLE 3

Ten rounds of +100% or −50%

Each round independently doubles or halves wealth with probability $\frac{1}{2}$. Find $\mathbb{E}W_{10}$, the median of W_{10}, and the chances of ending above $\mathbb{E}W_{10}$ and below W_0.

SOLUTION

With j wins, $W_{10}/W_0 = 2^{2j-10}$, and $j \sim \text{Bin}(10, \frac{1}{2})$. Each round multiplies expected wealth by $\frac{1}{2}(2 + 0.5) = 1.25$, and independent factors multiply in expectation, (16.1), so $\mathbb{E}W_{10}/W_0 = 1.25^{10} = 9.3132$.

Five wins return exactly to the start: $\Pr(W_{10} = W_0) = 63/256 = 0.2461$. Wealth is at most W_0 with probability $638/1{,}024$, and at least W_0 with the same probability, so the median is W_0.

Only $j \geq 7$ ends above the mean, since $2^4 = 16 > 9.3132 > 2^2$. Those paths have probability $11/64 = 0.1719$ but carry 8.1875 of the 9.3132, 87.9% of the mean (Exhibit 3). With $j \leq 4$ the investor loses: $\Pr(W_{10} < W_0) = 193/512 = 0.3770$. After 100 rounds, $\mathbb{E}W_{100}/W_0 = 4.91 \times 10^9$, yet $\Pr(W_{100} > \mathbb{E}W_{100}) = 0.00044$ and $\Pr(W_{100} < W_0) = 0.4602$.

Exhibit 3: Ten rounds of +100% or −50%: where the mean comes from

Wins j	W_{10}/W_0	Paths	Probability	Contribution to $\mathbb{E}W_{10}/W_0$
0	1/1,024	1	0.0010	0.0000
1	1/256	10	0.0098	0.0000
2	1/64	45	0.0439	0.0007
3	1/16	120	0.1172	0.0073
4	1/4	210	0.2051	0.0513
5	1	252	0.2461	0.2461
6	4	210	0.2051	0.8203
7	16	120	0.1172	1.8750
8	64	45	0.0439	2.8125
9	256	10	0.0098	2.5000
10	1,024	1	0.0010	1.0000
Total		1,024		9.3132

Note: paths $\binom{10}{j}$; probability = paths/1,024; contribution = paths/1,024 × W_{10}/W_0, rounded to four decimals.

Peters (2019) makes the same point with a coin paying +50% or −40%. Expected wealth grows by a factor 1.05 a round. Expected log wealth changes by $\frac{1}{2}\ln 0.9 = -0.0527$ a round, a typical factor of $\sqrt{0.9} = 0.949$.

Realized growth: the median path

Over m days, by (1.12),

$$\ln \frac{W_m}{W_0} = \sum_{t=1}^{m} \ln(1 + R_t) = \sum_{t=1}^{m} r_t,$$

so the strong law applies to the average log return.

THEOREM **Realized growth rate**

Let the daily returns $R_1, R_2, \ldots$ be iid with $1 + R_t > 0$ and $\mathbb{E}|\ln(1 + R_t)| < \infty$, and let $g = \mathbb{E}[\ln(1 + R)]$. Then, with probability one,

$$\frac{1}{m} \ln \frac{W_m}{W_0} \to g. \tag{17.3}$$

For LM4's binary bet at a position f, the log return is $\ln(1+fb)$ with probability p and $\ln(1-fa)$ with probability q, so (4.7) is the two-outcome case of (17.3).

For two positions with $g(f) > g(\tilde{f})$ on the same trades, the average log growths differ in the limit by $g(f) - g(\tilde{f}) > 0$. The ratio of their wealths thus grows without bound, with probability one (Breiman 1961; Thorp 2006). General bets, and what maximizing the median path assumes, are LM41's and LM42's subjects.

Expected wealth: the mean path

When the R_t are independent with mean μ and $\mathbb{E}|R_t| < \infty$, the factors multiply in expectation by (16.1):

$$\mathbb{E}W_m = W_0 \prod_{t=1}^{m} \mathbb{E}(1+R_t) = W_0(1+\mu)^m.$$

Call $W_0(1+\mu)^m$ the **mean path** and $W_0 e^{mg}$ the **median path**; Section 6 shows that the latter is approximately the median of W_m. By strict Jensen, (14.10) and (14.12), $g < \ln(1+\mu)$ unless R is constant. The difference $\delta := \ln(1+\mu) - g > 0$ is LM14's daily Jensen gap, and over m days the mean path pulls ahead by the factor $e^{m\delta}$.

THEOREM **The mean path runs away from the median path**

Let $R_1, \ldots, R_m$ be independent with mean μ and $\mathbb{E}|R_t| < \infty$. Then $\mathbb{E}W_m = W_0(1+\mu)^m$. If, in addition, the R_t are iid with $1 + R_t > 0$ and $0 < \sigma_r < \infty$, then

$$\Pr\left(W_m > \mathbb{E}W_m\right) \to 0 \qquad (m \to \infty). \tag{17.4}$$

Proof of the limit. The event $\{W_m > \mathbb{E}W_m\}$ says that the average log return $\frac{1}{m}\ln(W_m/W_0)$ exceeds its mean g by more than δ, which is positive because $\sigma_r > 0$. By (17.2) its probability is at most a constant divided by m (Problem 8). ■

Expected wealth is carried by rare paths far above the median, as in Exhibit 3; the iid bootstrap of LM18 has exactly this mean path.

EXAMPLE 4

Resolving the case: compounding the average day

TR23 has $\bar{R} = 0.062505\%$ and $\bar{r} = 0.055170\%$ a day. Explain the gap between the mean path's 38.66 and the realized 25.20. Then estimate the chance of ending above the mean path after $m = T$ days, in an iid model that draws days from the sample.

SOLUTION

The mean path is $(1+\bar{R})^{5849} = 38.66$, and the realized wealth is $e^{T\bar{r}} = 25.20$ by (1.12), the fitted median path. LM14 found the daily Jensen gap $\delta = \ln(1+\bar{R}) - \bar{r} = 0.7316$ bp, which needs six significant figures: $\bar{R} = 0.0625051\%$, $\bar{r} = 0.0551696\%$. Over the sample, $5{,}849 \times 0.7316 \text{ bp} = 0.4279$, and $e^{0.4279} = 1.534 = 38.66/25.20$.

With $\sigma_r = s_r = 1.2107\%$, Section 6 justifies the normal approximation

$$\begin{aligned} \Pr(W_m > \mathbb{E}W_m) &\approx \Phi\big(-\sqrt{m}\,\delta/\sigma_r\big), \\ \sqrt{5849} \times \frac{0.7316 \text{ bp}}{1.2107\%} &= 0.462, \\ \Phi(-0.462) &= 0.32. \end{aligned}$$

After 23.8 years the chance is still 0.32. A chance of 5% needs $\sqrt{m}\,\delta/\sigma_r \geq 1.645$: about 74,000 days, or 300 years. Chebyshev's bound, $\sigma_r^2/(T\delta^2) = 4.68$, is empty.

EXAMPLE 5

Leverage on TAIEX

Hold a constant leverage f on TR23's days, rebalanced daily with free borrowing and no costs (a hypothetical). Compare realized wealth with the mean path $(1+f\bar{R})^T$ for f = 1, 2, 4, 6 and 8, and find the growth-maximizing leverage.

SOLUTION

Realized wealth is $W_T/W_0 = e^{T\bar{g}(f)}$, and annual log growth $A\bar{g}(f) = \ln(W_T/W_0)/23.7645$:

Leverage f	W_T/W_0	$A\bar{g}(f)$	Mean path $(1+f\bar{R})^T$
1	25.20	13.58%	38.66
2	267.7	23.52%	1,491
4	2,127.8	32.24%	2.20×10^6
6	405.6	25.27%	3.23×10^9
8	1.175	0.68%	4.68×10^{12}

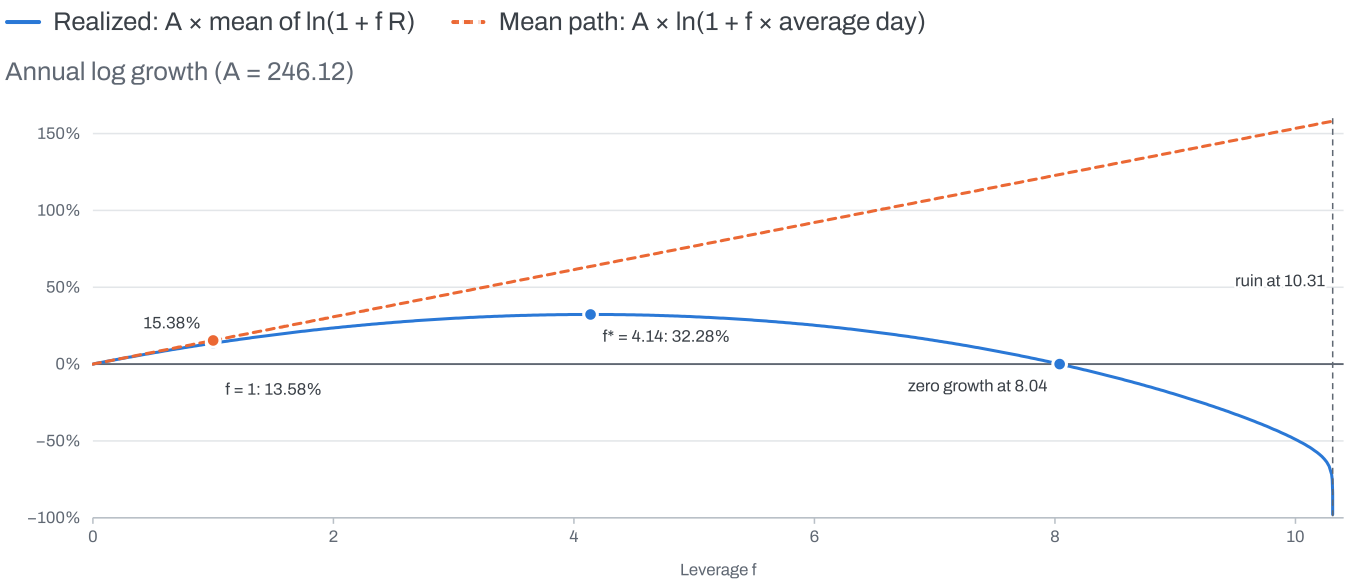
The mean path rises with every f; realized wealth peaks. The root of $\bar{g}'(f) = 0$, (4.5), is $f^* = 4.1381$: 32.28% a year, 2,147 times the start and 85 times the unlevered end. Growth is zero again at 8.04, or 1.94 times f^*, LM4's "about twice" (Exhibit 4).

Leverage makes the mean path hard to beat. At the Kelly leverage, $\delta(f^*) = 12.72$ bp and $s_r(f^*) = 5.079\%$, so after 5,849 days

$$z = \sqrt{5849} \times \frac{12.72 \text{ bp}}{5.079\%} = 1.915,$$
$$\Pr(W_m > \mathbb{E}W_m) \approx \Phi(-z) = 0.028.$$

The realized path is hard to live through. At f^* it fell 98.5% from its peak by 20 November 2008, a risk LM41 returns to.

Exhibit 4: Annual log growth of constant-leverage TAIEX: realized and mean path



Note: hypothetical daily-rebalanced leverage on TR23, free borrowing, no costs, $A = 246.12$. Realized $A\bar{g}(f)$ is LM14's 13.58% at $f = 1$; the mean path's $A\ln(1 + f\bar{R})$ is 15.38%. The realized curve peaks at $f^* = 4.14$ (32.28%), crosses zero at 8.04 and falls to $-\infty$ at $1/|R_{(1)}| = 10.31$, where $R_{(1)} = -9.6997\%$ (7 April 2025). Source: FinMind `TaiwanStockTotalReturnIndex` (TAIEX), retrieved 18 October 2026.

PITFALL **The law fixes the rate, not the level**

Formula (17.3) says that $\frac{1}{m}\ln(W_m/W_0)$ settles at g, not that W_m settles near W_0e^{mg}. The log of their ratio has standard deviation $\sigma_r\sqrt{m}$, which grows. Over TR23's 5,849 days $s_r\sqrt{T} = 0.9259$, so one standard deviation is a factor $e^{0.9259} = 2.52$ either way.

導讀 **期望值路徑與中位數路徑**

財富取對數後是每日對數報酬（log return）的和，所以依強大數法則（strong law of large numbers），實現的每日對數成長率會收斂到 $g = \mathbb{E}[\ln(1 + R)]$。期望值（expectation）卻以 $\ln(1 + \mu)$ 的速度複利，比 g 快，差距 δ 來自 Jensen 不等式（Jensen's inequality）。期望值靠少數極端好的路徑撐起來，一般投資人走的是中位數（median）路徑 W_0e^{mg}。期數拉長，贏過期望值路徑的機率會趨近於零，但很慢：用加權股價報酬指數的獨立同分布（iid）模型，23.8 年後仍有 0.32。加上 Kelly 槓桿就快得多，只剩約 0.028，代價是那條路徑在 2008 年一度跌掉 98.5%。

KNOWLEDGE CHECK 3

Trade LM3's +2%/−1% bet 1,000 times at full stake, with LM3's $g = 0.48761\%$ per trade. Compute the mean path and the median path.

ANSWER

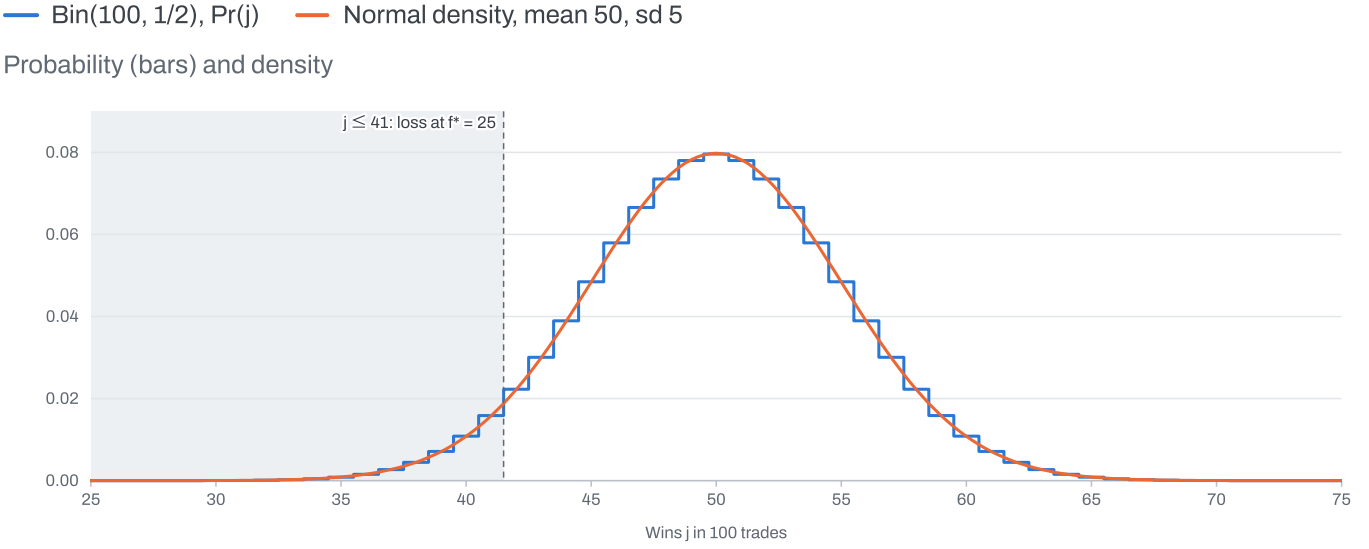
The mean path is $1.005^{1000} = 146.58$ times the start, and the median path $e^{1000 \times 0.0048761} = 131.12$ times.

5 THE CENTRAL LIMIT THEOREM

- ☐ state the central limit theorem with its conditions, and apply the normal approximation to binomial counts

The central limit theorem says how an average is spread on its way to the mean. LM4 held LM3's +2%/−1% bet at its Kelly position $f^* = 25$. Call this the Kelly bet: each trade multiplies wealth by 1.5 or 0.75 with equal probability, with growth $g^* = 5.889\%$ per trade. After 100 trades the number of wins is $\text{Bin}(100, \frac{1}{2})$. Its histogram in Exhibit 5 is close to the normal density with mean 50 and standard deviation 5.

Exhibit 5: Wins in 100 trades of the Kelly bet: binomial and normal



Note: $\Pr(j)$ for $j \sim \text{Bin}(100, \frac{1}{2})$, each over $[j - \frac{1}{2}, j + \frac{1}{2}]$, and the density $\varphi\big((x - 50)/5\big)/5$. At $f^* = 25$ wealth falls below W_0 exactly when $j \le 41$ (shaded): 0.04431 exactly, 0.04464 by the normal approximation, 0.04457 with the continuity correction (Example 6).

DEFINITION **Convergence in distribution**

The standardized sums $Z_n := (S_n - n\mu_X)/(\sigma_X\sqrt{n})$ converge **in distribution** to a standard normal Z when $\Pr(Z_n \le z) \to \Phi(z)$ for every z. In general, distribution functions must converge wherever the limit's is continuous.

THEOREM **Central limit theorem**

Let $X_1, X_2, \ldots$ be iid with mean μ_X and variance $0 < \sigma_X^2 < \infty$. Then, as $n \to \infty$, for every real z

$$\Pr\left(\frac{S_n - n\mu_X}{\sigma_X\sqrt{n}} \le z\right) \to \Phi(z). \tag{17.5}$$

Stated without proof (Wasserman 2004, Theorem 5.8; Grinstead and Snell 1997, §9.1–§9.3).

So for large n the average is approximately $\mathcal{N}(\mu_X, \sigma_X^2/n)$. The theorem gives no error for a given n, and error bounds are not in this book.

DEEP DIVE **Why sums become normal** optional · click to expand

Here t is a real argument, and $M_X(t) := \mathbb{E}[e^{tX}]$ is the **moment generating function** (mgf), assumed finite near 0. For a centred draw $Y = X - \mu_X$, expanding e^{tY} by (3.2) term by term (stated without proof) gives

$$M_Y(t) = 1 + \tfrac{1}{2}\sigma_X^2 t^2 + o(t^2) \qquad (t \to 0),$$

where $o(t^2)/t^2 \to 0$. Independent terms multiply their mgfs, by (16.1), so

$$\begin{aligned} M_{Z_n}(t) &= \left[M_Y\left(\frac{t}{\sigma_X\sqrt{n}}\right)\right]^n \\ &= \left[1 + \frac{t^2}{2n} + o\left(\frac{1}{n}\right)\right]^n \to e^{t^2/2}. \end{aligned}$$

The limit extends (1.6) to $x_n \to x$ by (3.3), and $e^{t^2/2} = \mathbb{E}[e^{tZ}]$ by (5.12). That converging mgfs give converging distribution functions is stated without proof (Grinstead and Snell 1997, §10.3; Blitzstein and Hwang 2019, Theorem 10.3.1).

A binomial count is a sum of n iid indicators, so it has mean np and variance $np(1-p)$, (15.2). For an integer j_0, (17.5) with the continuity correction gives

$$\Pr(j \le j_0) \approx \Phi\left(\frac{j_0 + \frac{1}{2} - np}{\sqrt{np(1-p)}}\right).$$

The $\frac{1}{2}$ is the **continuity correction**: the bar of j_0 spans $[j_0 - \frac{1}{2}, j_0 + \frac{1}{2}]$, so the normal area should stop at $j_0 + \frac{1}{2}$.

EXAMPLE 6

The chance of being behind at full Kelly

Hold the Kelly bet ($f^* = 25$) for $n = 100$ trades. Find $\Pr(W_n < W_0)$ exactly, by the normal approximation of the log growth and with the continuity correction. Repeat for $n = 10$.

SOLUTION

The log growth $j \ln 1.5 + (n - j) \ln 0.75$ is negative exactly when $j < n \ln(4/3)/\ln 2 = 0.4150375\, n$ (LM4's 0.41504). At $n = 100$ that means $j \le 41$, with exact probability 0.04431.

The per-trade log return has mean $g^* = 0.0588915$ and standard deviation $\sigma_r = \ln 2/2 = 0.3465736$. By (17.5),

$$\begin{aligned} \Pr(W_n < W_0) &\approx \Phi\big(-g^*\sqrt{n}/\sigma_r\big) \\ &= \Phi(-1.69925) \\ &= 0.04464. \end{aligned}$$

The value $z = -1.69925$ keeps six significant figures because 0.04464 lies only 1.0×10^{-6} above a rounding boundary. Corrected, $\Phi((41.5 - 50)/5) = \Phi(-1.7) = 0.04457$.

At $n = 10$ the corrected normal, 0.37591, is close to the exact 0.37695, but the uncorrected one gives 0.29551.

EXAMPLE 7

One percentage point with 95%

Redo Example 2 with the normal approximation.

SOLUTION

By (17.5), j/n is approximately normal with standard deviation $\sqrt{p(1-p)/n} \le 0.5/\sqrt{n}$. A miss beyond 0.01 has probability about 5% when $0.01 = 1.96 \times 0.5/\sqrt{n}$, with LM4's 1.96. So $n \ge 1.96^2 \times 0.25/0.01^2 = 9{,}604$, against Chebyshev's 50,000. This, too, is a property of a model, not an interval for a data frequency (LM20).

PITFALL **The theorem is about the centre, not the far tail**

Far in the tail a small absolute error can be a large relative one. After 400 trades of the Kelly bet the exact chance of being behind is about 0.0004. The continuity-corrected normal approximation is 2.6% too high, against 0.6% at 100 trades, and the uncorrected one is 14% too low (Problem 6 explains why). Check a normal approximation near 10^{-4} against an exact value or a simulation (LM18).

導讀 **中央極限定理：和會變成常態**

中央極限定理（central limit theorem）說：獨立同分布（iid）的抽樣只要變異數（variance）有限，標準化之後的和會依分布收斂（convergence in distribution）到標準常態分布（standard normal distribution）。所以 n 夠大時，和與平均數（mean）都「近似」常態；這是極限定理，並沒有說某個 n 的誤差多大。二項分布（binomial distribution）的計數是 n 個指示函數（indicator）的和，常態逼近（normal approximation）加上連續性校正（continuity correction），把面積算到 $j_0 + \frac{1}{2}$，通常相當準。在遠端尾部，絕對誤差雖小，相對誤差卻可能很大。Kelly 賭局 400 次後落後的機率約 0.0004：加上連續性校正的常態逼近高估 2.6%（100 次時只高估 0.6%），未校正的則低估 14%。

KNOWLEDGE CHECK 4

Estimate the probability of at least 60 heads in 100 tosses of a fair coin with the continuity correction, and compare it with the exact 2.84%.

ANSWER

Here $\Pr(j \geq 60) = 1 - \Pr(j \leq 59) \approx 1 - \Phi\big((59.5 - 50)/5\big) = 1 - \Phi(1.9) = 2.87\%$, against the exact 2.84%.

6 WEALTH IS APPROXIMATELY LOGNORMAL

- ☐ compute the median, quantiles and loss probability of wealth under the lognormal approximation, with and without leverage

By (17.5), the sum $\ln(W_m/W_0)$ of m iid log returns is approximately $\mathcal{N}(mg, m\sigma_r^2)$, so W_m/W_0 is approximately lognormal, (15.7). The exponential is increasing, so the quantiles of W_m are the exponentials of the normal quantiles of (15.6).

RESULT **Lognormal approximation of wealth**

Let the daily returns be iid with $1 + R > 0$ and $0 < \sigma_r < \infty$, and let m be large. Then

$$\begin{aligned} \Pr(W_m < W_0) &\approx \Phi\Big(- \frac{g\sqrt{m}}{\sigma_r} \Big), \\ F^{-1}_{W_m}(u) &\approx W_0 e^{mg + \Phi^{-1}(u)\,\sigma_r \sqrt{m}}, \end{aligned} \tag{17.6}$$

so the median is about $W_0 e^{mg}$ and the 5% and 95% quantiles about $W_0 e^{mg \mp 1.645\,\sigma_r\sqrt{m}}$. Each "$\approx$" is the error of (17.5), large in relative terms far in the tail. For iid normal log returns, (17.6) is exact.

Likewise $\{W_m > \mathbb{E}W_m\}$ is $\{\ln(W_m/W_0) - mg > m\delta\}$, so $\Pr(W_m > \mathbb{E}W_m) \approx \Phi(-\sqrt{m}\,\delta/\sigma_r)$, as used in Example 4.

LM5 assumed that 0050's annual log growth is normal, with median growth factor 1.2130 and mean 1.2358. The central limit theorem is why that can work: a year's log growth sums about 243 daily log returns, which should be iid with a finite variance. LM38 derives the lognormal law exactly for geometric Brownian motion, into which more and smaller steps turn this random walk (LM36).

The lognormal mean $e^{m(g+\sigma_r^2/2)}$ of (5.12) can miss the exact $(1+\mu)^m$ badly, because a mean depends on the far tail.

EXAMPLE 8

One year, ten years and the whole sample

Take the iid model with TR23's empirical law, in-sample: $g = \bar{r} = 0.055170\%$ and $\sigma_r = s_r = 1.2107\%$ a day. Find the chance of a loss over one year ($m = 246$) and ten years ($m = 2{,}461$), and the ten-year median and 5% and 95% quantiles. Compare one year with the calendar years.

SOLUTION

One year: $z = 0.055170\% \times \sqrt{246}/1.2107\% = 0.7147$, so $\Pr(W_{246} < W_0) \approx \Phi(-0.7147) = 23.74\%$. Ten years: $z = 2.261$ and 1.19%, with median $e^{2461\bar{r}} = 3.89$ and quantiles 1.45 and 10.44. Over the sample the median is 25.20 by construction and the 90% range 5.49 to 115.6 (Exhibit 6).

LM15's lognormal, fitted to the 23 calendar-year growth factors of 2003–2025, loses in 30.0% of years. The daily model refitted to those years' 5,664 returns gives, by (17.6), 26.54%, or 6.10 losing years; 5 lost (LM15).

The means nearly agree, 11.634% (246 times the refitted daily mean, 0.047292%) against 11.646% a year, as both spread the same total log growth. The gap is dispersion. Annual log returns have a standard deviation of 22.207%, while the refitted daily standard deviation, 1.18352%, gives only $\sqrt{246} \times 1.18352\% = 18.563\%$, and $\Phi(-11.634/18.563)$ gives the 26.54%. The full-sample 23.74% is lower because the index rose 73.0% in 2026's nine months. LM19 examines the $\sqrt{m}$ scaling.

Exhibit 6: The lognormal approximation of TAIEX wealth

Horizon m (sessions)	Median	Mean path	5% quantile	95% quantile	$\Pr(W_m < W_0)$	$\Pr(W_m > \mathbb{E}W_m)$
One year, 246	1.1454	1.1662	0.8381	1.5653	23.74%	46.2%
Ten years, 2,461	3.89	4.65	1.45	10.44	1.19%	38.2%
Sample (23.76 years), 5,849	25.20	38.66	5.49	115.6	0.025%	32.2%

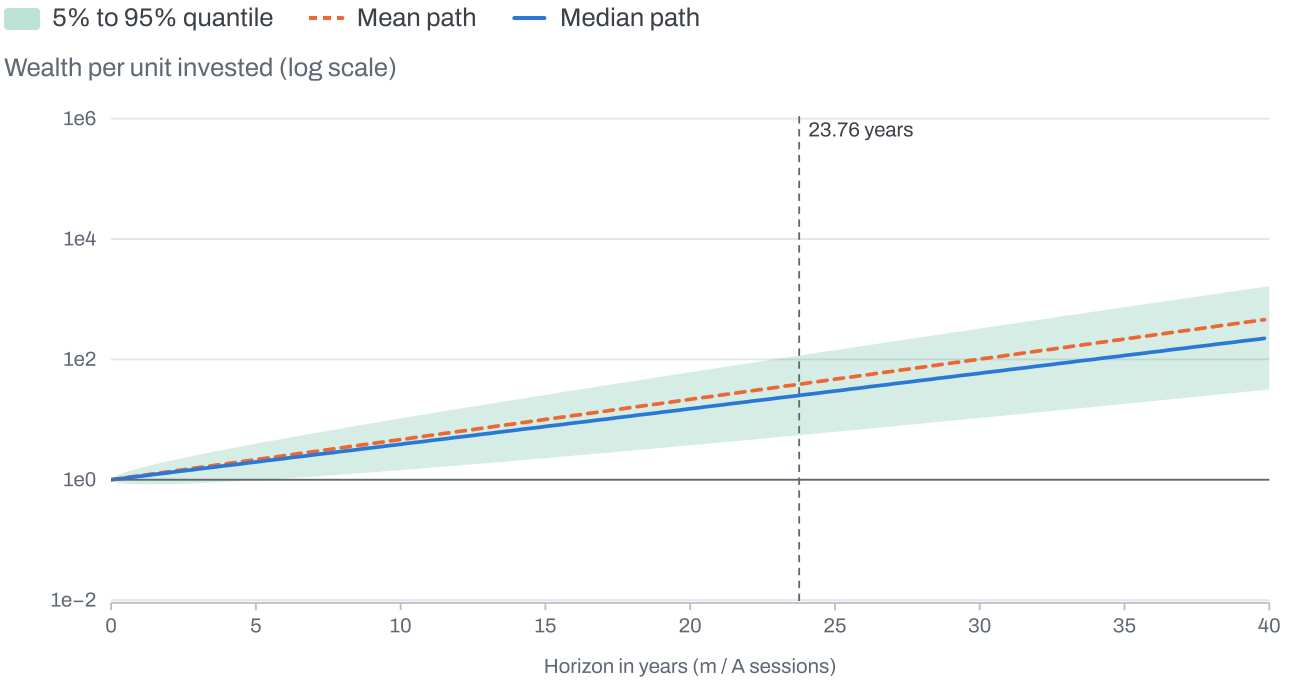
Note: W_m/W_0 in the iid model on TR23's empirical law at $f = 1$, in-sample, with $\bar{R} = 0.062505\%$, $\bar{r} = 0.055170\%$ and $s_r = 1.2107\%$. Median $e^{m\bar{r}}$, mean path $(1+\bar{R})^m$; the rest by (17.6), the normal approximation, and for $\Pr(W_m > \mathbb{E}W_m)$ by its form in Section 6. The model's exact values (the exact law of the sum, computed numerically) differ by at most 0.2 points for the probabilities and 0.2% for the quantiles. The exact loss probability is 23.62% at one year and 0.026% over the sample.

Descriptive: 5 of the 23 calendar years 2003–2025 (2003 from its first close, 2 January 2003) lost, namely 2008, 2011, 2015, 2018 and 2022. For the daily model refitted to them, (17.6) gives 6.10 losing years (Example 8). Source: FinMind `TaiwanStockTotalReturnIndex` (TAIEX), retrieved 18 October 2026.

The widget draws (17.6) for any leverage and horizon, starting from the sample row of Exhibit 6. At five years, $m = 1{,}231$, it reads a 5.49% chance of a loss, LM18's case.

Mean, median and the range of wealth

INTERACTIVE



Mean path, exact iid mean	Median path	5% and 95% quantiles
38.66	**25.20**	**5.49 · 115.6**
Chance of ending below the start	**Chance of ending above the mean path**	**Horizon: sessions · years**
0.025%	**32.20%**	**5,849 · 23.76**

Model: days drawn independently from TAIEX's 5,849 daily total returns, in-sample, not a forecast; leverage is hypothetical (daily rebalancing, free borrowing, no costs). The median, the 5% to 95% band and both chances use the lognormal approximation (17.6) with the mean and standard deviation of ln(1 + f R) at that leverage. The mean path is the exact iid mean, (1 + f × average day)^m; the lognormal mean would give 38.69 instead of 38.66 over the sample at f = 1, and 2.19e13 instead of 4.68e12 at f = 8.

PITFALL **A fitted median that matches the sample is not evidence**

The median path $e^{T\bar{r}}$ ends exactly at TR23's realized 25.20, because $T\bar{r} = \ln(W_T/W_0)$ by (1.12). The match is arithmetic, not a test of the iid model. Nor does the path that fitted the model test its range of 5.49 to 115.6.

導讀 **財富近似對數常態分布**

財富取對數是許多日對數報酬（log return）的和，依中央極限定理（central limit theorem）近似常態，所以財富本身近似對數常態分布（lognormal distribution）。中位數（median）約為 $W_0 e^{mg}$，分位數（quantile）就是常態分位數取指數，虧損機率約為 $\Phi(-g\sqrt{m}/\sigma_r)$。在加權股價報酬指數的獨立同分布（iid）模型下，一年虧損的機率約 23.74%，十年降到 1.19%；2003–2025 年的 23 個日曆年中，實際有 5 年虧損。樣本內配適的中位數路徑剛好落在實際的 25.20 倍，這是算術恆等式，不是模型的證據。期望值（expectation）由遠端尾部撐起，而常態逼近（normal approximation）在那裡最不準。

KNOWLEDGE CHECK 5

At leverage $f = 2$, TR23's $\ln(1 + 2R_t)$ has mean $\bar{g}(2) = 0.095571\%$ and standard deviation $s_r(2) = 2.4288\%$. Under the iid model, what is the chance of a loss over one year, $m = 246$? Compare it with $f = 1$.

ANSWER

By (17.6), $z = 0.095571\% \times \sqrt{246}/2.4288\% = 0.6172$, so $\Pr(W_{246} < W_0) \approx \Phi(-0.6172) = 26.86\%$, against 23.74% at $f = 1$. Leverage raises the median growth but raises the spread more.

7 HOW FAST THE NORMAL ARRIVES ON TAIEX

- [ ] measure how fast sums of TAIEX log returns approach the normal shape, and explain how dependence slows the approach

A sum of fat-tailed iid days becomes normal only as m grows. Excess kurtosis, Kurt -3 by (14.5), measures the progress: it is 0 for a normal law by (14.6).

Let $X_1, \dots, X_m$ be iid with $\mathbb{E}X^4 < \infty$ and $\sigma_X > 0$, and let $Y_i = X_i - \mu_X$. In the expansion of $(Y_1 + \cdots + Y_m)^4$, a product in which some index appears once has expectation 0, by (16.1) and $\mathbb{E}Y_i = 0$. The m products Y_i^4 have expectation $\mathrm{Kurt}(X)\,\sigma_X^4$ each, by (14.5), and the $3m(m-1)$ products $Y_i^2 Y_k^2$ with $i \neq k$ have σ_X^4 each (Problem 9). Dividing by $(\mathrm{Var}\, S_m)^2 = m^2\sigma_X^4$,

$$\begin{aligned} \mathrm{Kurt}(S_m) &= \frac{m\,\mathrm{Kurt}(X) + 3m(m-1)}{m^2}, \\ \mathrm{Kurt}(S_m) - 3 &= \frac{\mathrm{Kurt}(X) - 3}{m}. \end{aligned} \tag{17.7}$$

Sample kurtosis and its divisor are LM19's subject; here the divisor is T, as in LM5.

EXAMPLE 9

TAIEX sums against the iid prediction

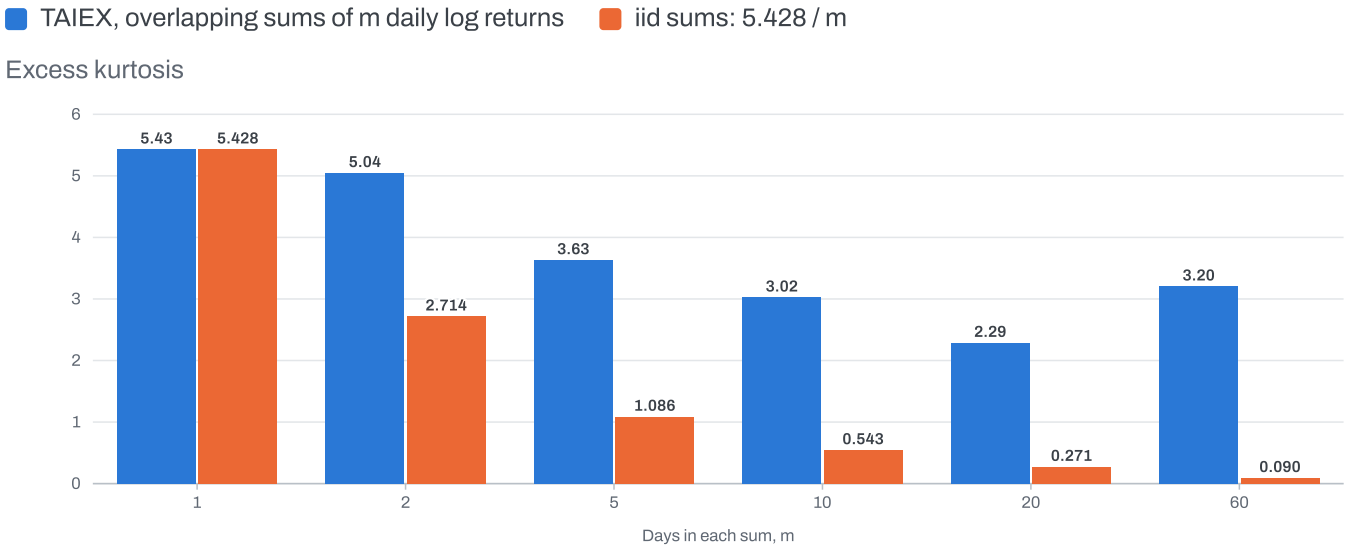
TR23's daily log returns have excess kurtosis 5.428. What does (17.7) predict for sums of 5 and 20 days, and what do TR23's overlapping sums show?

SOLUTION

The prediction is $5.428/5 = 1.09$ and $5.428/20 = 0.27$. TR23's sums over all windows of 5 and 20 consecutive days show 3.63 and 2.29 (Exhibit 7): the normal shape arrives far more slowly.

The cause is dependence between squares. Next-day log returns correlate only 0.039, but their absolute values correlate 0.161. The pairs of absolute values twenty days apart still correlate 0.123. When large moves cluster, the terms $\mathbb{E}[Y_i^2 Y_k^2]$ exceed σ_X^4 and raise the kurtosis of the sum. LM26 models such changing volatility.

Exhibit 7: Excess kurtosis of TAIEX log-return sums over m days, against iid sums



Note: TR23's overlapping sums of m daily log returns, all $5{,}849 - m + 1$ windows; the iid bars are $5.428/m$ by (17.7). At $m = 1$ both bars are the daily value (simple returns: LM14's 5.27). Non-overlapping windows depend on the starting day: 0.69 to 4.49 for 20 days (2.40 from the first return, 292 windows), 0.42 to 7.22 for 60 days (2.76, 97 windows). The rise from 2.29 to 3.20 is thus evidence neither way.

Stability check: with windows wholly within each regime, the 20-day value is 1.96 before 1 June 2015 and 2.66 after. The iid values are 0.17 and 0.42, from the regimes' daily 3.44 and 8.41. Source: FinMind `TaiwanStockTotalReturnIndex` (TAIEX), retrieved 18 October 2026.

PITFALL **Overlapping windows are not independent sums**

TR23's 5,830 overlapping 20-day sums share 19 days with their neighbours: they are far from 5,830 independent observations. Non-overlapping sums are few and depend on the starting day. State the window type with any kurtosis of long sums.

KNOWLEDGE CHECK 6

Daily TR23 log returns have excess kurtosis 5.428. For iid days, how many days must a sum contain before its excess kurtosis falls below 0.5? What does TR23 show at 10 days?

ANSWER

By (17.7) we need $5.428/m < 0.5$, that is $m > 10.856$, so 11 days. TR23's overlapping 10-day sums show 3.02, more than five times the iid 0.543.

8 WHEN THE LAWS FAIL

- [ ] identify when the laws fail (no mean, infinite variance, infinite fourth moment, dependence) and what each failure does to a sample statistic

Exhibit 8 lists what happens when a condition is missing, for the laws of LM5's tails exhibit and for dependent draws.

Exhibit 8: What each law needs, and what happens without it

Draws	Mean	Variance	Fourth moment	Law of large numbers	Normal limit with $\sigma_X/\sqrt{n}$	Sample kurtosis settles
Normal	finite	finite	finite	yes	yes (exact)	yes
Student t_3	finite	finite	infinite	yes	yes	no
Pareto, $\alpha = 1.5$	finite	infinite	infinite	yes (strong law)	no	no
Cauchy (t_1)	none	none	none	no	no	no
Identical copies $X_i = X_1$	as X_1	as X_1	as X_1	no	no	undefined

Note: iid draws in the first four rows. The law of large numbers needs a finite mean, (17.5) a finite variance, and a settling kurtosis a finite fourth moment. Identical copies give constant samples, with no kurtosis.

No mean. The standard Cauchy density $1/\big(\pi(1+x^2)\big)$ is the t density (15.9) with $\nu = 1$. Its tails fall like x^{-2}, so $\mathbb{E}|X| = \infty$ and the mean does not exist (LM5).

EXAMPLE 10

Averages that never settle

Compare $\Pr(|\bar{X}_n| > 1)$ for averages of n independent standard Cauchy draws and of standard normal draws, at $n = 1$, 4 and 25.

SOLUTION

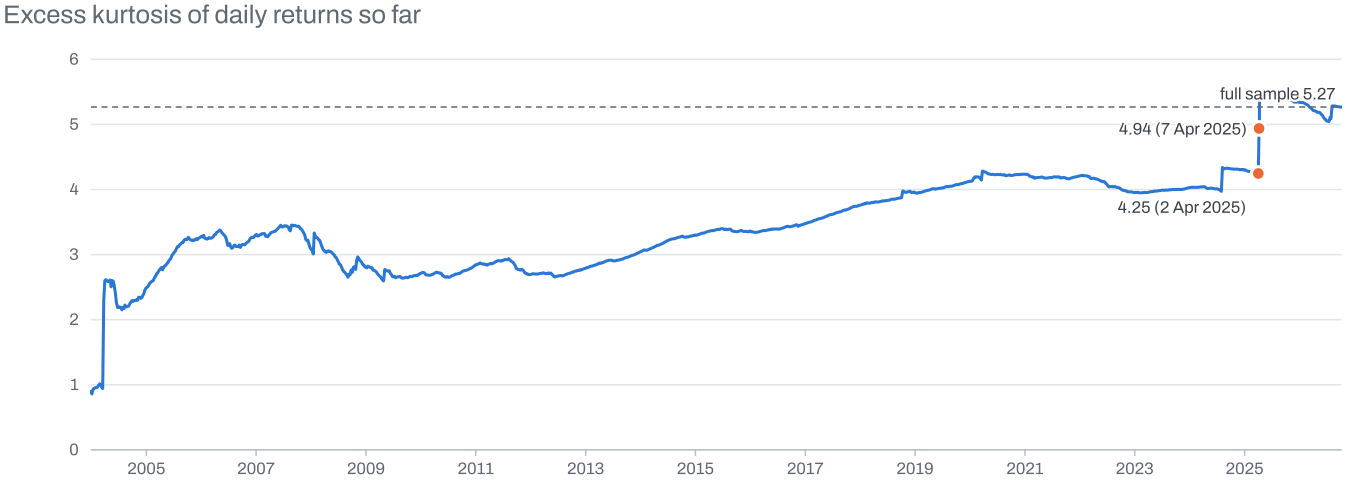
The average of n independent standard Cauchy draws is again standard Cauchy (stated without proof; Grinstead and Snell 1997, §8.2, Example 8.8). Its distribution function is $\frac{1}{2} + \frac{1}{\pi}\arctan x$, so $\Pr(|\bar{X}_n| > 1) = 1 - \frac{2}{\pi}\arctan 1 = \frac{1}{2}$ for every n. A normal average has standard deviation $1/\sqrt{n}$, so the chance is $2\Phi(-\sqrt{n})$: 0.317, 0.0455 and 5.7×10^{-7}. Without a finite mean, iid averages converge with probability zero (Durrett 2019, Theorem 2.3.8).

No variance. A Pareto draw with $\alpha = 1.5$, or a t draw with $1 < \nu \leq 2$, has a mean but no variance. The strong law holds, but sums need a larger scale than $\sqrt{n}$ and tend to **stable** laws (stated without proof; that theorem is not in this book). Mandelbrot (1963) modelled cotton prices with such laws; Cont (2001) puts the tail index of most return series between 2 and 5.

No fourth moment. With a tail index near 3, as LM5 reported for index returns, the mean and the variance exist but the fourth moment does not. The running mean and variance settle, while the sample kurtosis estimates an infinite quantity. Exhibit 9 shows TR23's running excess kurtosis, which 7 April 2025 alone moved from 4.25 to 4.94 (LM15).

A few days decide it. The ten largest of TR23's 5,849 terms supply 35.0% of $\sum_t (R_t - \bar{R})^4$ but only 6.7% of $\sum_t (R_t - \bar{R})^2$ (LM15). For 0050, LM5 found 62.8% and 14.1%, and a jump of the excess kurtosis from 5.84 to 7.63 on the same day.

Exhibit 9: Running excess kurtosis of TAIEX daily returns



Note: running excess kurtosis of TR23's daily simple returns, divisor the days so far, from the 246th return (29 December 2003). The line marks LM14's full-sample 5.27. The running annualized variance of R (same divisor, times A) stays between 0.0326 (12 April 2024) and 0.0580 (23 June 2004), ending at 0.0359. On 7 April 2025 it rose 1.3% (0.03335 to 0.03377) while the excess kurtosis rose 16%: the second moment settles, the fourth does not. Source: FinMind `TaiwanStockTotalReturnIndex` (TAIEX), retrieved 18 October 2026.

Dependence and change. Identical copies are the extreme of dependence. Milder dependence, such as Section 7's clustering, slows the approach to the normal shape. Laws for dependent sequences, the ergodic theorems, are not in this book. A changing distribution also breaks iid: TR23's daily log returns had excess kurtosis 3.44 before 1 June 2015 and 8.41 after, when the daily limit was 10% instead of 7%.

PITFALL **A finite sample kurtosis shows no finite fourth moment**

Every sample has a finite kurtosis: TR23's excess kurtosis is 5.27 (kurtosis 8.27, LM14), whatever the population. What signals trouble is a running value that jumps with single days, as in Exhibit 9. Quote a sample kurtosis with its period, and prefer tail measures that need fewer moments (LM15).

導讀 **定理失效的時候**

這些定理都有條件。沒有平均數（mean）時，例如 Cauchy 分布，樣本平均數（sample mean）永遠不會收斂：n 個獨立（independent）Cauchy 抽樣的平均仍是 Cauchy 分布。變異數（variance）無限大時，強大數法則（strong law of large numbers）仍然成立，但以 $\sqrt{n}$ 標準化的中央極限定理（central limit theorem）失效，極限是穩定分布（stable law）。四階動差（fourth moment）無限大時，樣本平均數與樣本變異數會收斂，樣本峰態（sample kurtosis）卻永遠穩定不下來；加權股價報酬指數的超額峰態（excess kurtosis）在 2025 年 4 月 7 日一天內從 4.25 跳到 4.94。獨立同分布（iid）的假設本身也可能不成立：大漲大跌會群聚，2015 年 6 月漲跌幅限制放寬以後，分布也變了。

KNOWLEDGE CHECK 7

Daily returns are iid with a power tail of index $\alpha = 3$. As the sample grows, which running statistics settle: the mean, the variance, the kurtosis?

ANSWER

The mean and the variance are finite, so by the strong law their running versions settle. The fourth moment is infinite, so the running kurtosis keeps jumping with extreme days. TR23's running excess kurtosis rose from 4.25 to 4.94 on 7 April 2025.

9 THE LAWS IN PYTHON

- [ ] compute running averages, sums over windows of m days and normal approximations in Python, counting threshold days with the book's guard

Log returns enter every sum, simple returns the count of up days. The guard `R > 1e-12` keeps a return that is zero up to rounding out of the up days (LM5).

PYTHON

```python
import numpy as np
import polars as pl
from scipy import stats

tr = pl.read_csv("data/taiex_tr.csv")["price"].to_numpy()
R = tr[1:] / tr[:-1] - 1                  # T = 5,849 simple returns
r = np.log1p(R)                           # log returns, for every sum
n = np.arange(1, len(R) + 1)
up_share = np.cumsum(R > 1e-12) / n       # running share of up days
mean_r = np.cumsum(r) / n                 # running mean log return


def window_sums(x, m):                    # all len(x) - m + 1 sums of m days
    c = np.concatenate([[0.0], np.cumsum(x)])
    return c[m:] - c[:-m]


for m in (1, 5, 20, 60):                  # 5.43, 3.63, 2.29, 3.20
    print(m, stats.kurtosis(window_sums(r, m)))
print(stats.binom.cdf(41, 100, 0.5))      # 0.04431, exact
print(stats.norm.cdf((41.5 - 50) / 5))    # 0.04457, continuity-corrected
```

By default `stats.kurtosis` returns excess kurtosis with divisor T (`fisher=True`, `bias=True`); Polars' `std` divides by $T - 1$ unless given `ddof=0`.

> **IN PYTHON** **Companion lab**
>
> The Polars-first notebook `LM17_lab.ipynb` rebuilds the module's exhibits and computational examples from TR23's closes, computes the widget's grid and solves the computational practice problems. It ends with the stability check of Exhibit 7 for each limit regime.
>
> The notebook reads FinMind files that you download once with your own token. From the repository root, run `uv run python data/fetch_finmind.py`, then `uv run python data/make_lab_data.py`. FinMind's licence does not allow the book to redistribute the data. After that, the notebook runs offline.

KNOWLEDGE CHECK 8

A colleague computes the running mean of the log returns as `np.cumsum(r) / np.arange(len(r))`. What goes wrong?

ANSWER

The divisor starts at 0, so the first running sum is divided by zero and every later one by $n-1$ instead of n. Use `np.arange(1, len(r) + 1)`.

SUMMARY

- An average of n uncorrelated draws keeps the mean and divides the variance by n, (17.1).
- Chebyshev's inequality gives the weak law (17.2); the strong law needs only a finite mean.
- Realized growth settles at g, (17.3); the mean path compounds faster, and the chance of beating it tends to zero, (17.4), slowly.
- Standardized sums are approximately normal when the variance is finite, (17.5), so wealth is approximately lognormal, (17.6).
- Excess kurtosis of iid sums falls like $1/m$, (17.7), but TAIEX's 20-day sums keep 2.29.
- Without a mean, a variance or a fourth moment, the matching law fails.

PRACTICE PROBLEMS

1. Four hundred independent trades each have a standard deviation of 1.2%. The standard deviation of their average return is *closest* to:

   A. 0.06%

   B. 0.003%

   C. 24%

2. A fair coin is tossed 2,500 times, with j heads. Chebyshev's bound (17.2) on $\Pr(|j/n - \tfrac{1}{2}| \geq 0.02)$ is *closest* to:

   A. 0.0455

   B. 0.5

   C. 0.25

3. A bet doubles or halves wealth with probability $\tfrac{1}{2}$ each, independently, for four rounds, so $\mathbb{E}W_4 = 1.25^4\, W_0 = 2.4414\, W_0$. The probability that $W_4 > \mathbb{E}W_4$ is:

   A. 1/16

B. 5/16

C. 11/16

4. Draws $X_1, X_2, \ldots$ are iid Student t with $\nu = 1.5$. As n grows, the average $\bar{X}_n$:

A. does not converge, because the variance is infinite.

B. is approximately normal with standard deviation $\sigma_X/\sqrt{n}$.

C. converges to μ_X with probability one, without a normal approximation of that form.

5. Daily log returns are iid with excess kurtosis 5.43. The excess kurtosis of their 20-day sums is *closest* to:

A. 1.21

B. 0.27

C. 5.43

6. Hold the Kelly bet ($f^* = 25$) for $n = 400$ trades. Compute $\Pr(W_{400} < W_0)$ exactly, by $\Phi(-g^*\sqrt{n}/\sigma_r)$ and with the continuity correction, to three significant figures. Why is the uncorrected value 14% too low here but 0.7% too high at $n = 100$?

7. In the iid model with TR23's empirical law ($\bar{r} = 0.055170\%$, $s_r = 1.2107\%$ a day, in-sample), take five years: $m = \text{round}(5A) = 1{,}231$ sessions with $A = 246.12$. Use (17.6) with $\Phi^{-1}(0.95) = 1.645$ for the chance of a loss (in %) and the median and the 5% and 95% quantiles of W_m/W_0, all to two decimals. Why is the 5% quantile below 1?

8. (a) Show from (17.2) that $\Pr(W_m > \mathbb{E}W_m) \le \sigma_r^2/(m\delta^2)$ under the conditions of (17.4). (b) At $f = 2$, TR23's daily Jensen gap is $\delta(2) = 2.936$ bp (LM14) and $s_r(2) = 2.4288\%$. In the iid model, evaluate the bound and $\Phi(-\sqrt{m}\,\delta(2)/s_r(2))$ at $m = 5{,}849$, to three significant figures.

9. (a) Show that the expansion of $(Y_1 + \cdots + Y_m)^4$ behind (17.7) has exactly $3m(m-1)$ of its m^4 products of the form $Y_i^2 Y_k^2$ with $i \neq k$. (b) A fair-coin indicator has kurtosis 1. Use (17.7) for the excess kurtosis of $\text{Bin}(m, \frac{1}{2})$, and evaluate it at $m = 20$.

10. The average of n independent standard Cauchy draws is standard Cauchy. (a) Show that $\Pr(|\bar{X}_n| > 1) = \frac{1}{2}$ for every n, and compare the normal value $2\Phi(-5)$ at $n = 25$. (b) Which step of the proof of (17.2) fails?

11. *(Python)* From `taiex_tr.csv`, compute TR23's running share of up days (`R > 1e-12`) after 246 and after 1,231 returns. Compute the excess kurtosis (`stats.kurtosis`, defaults) of overlapping sums of m daily log returns for m = 1, 5, 20 and 60, against $5.428/m$. Give the shares to four decimals and the excess kurtoses to two.

SOLUTIONS

1. **A** is correct: $1.2\%/\sqrt{400} = 0.06\%$ by (17.1). B divides by n instead of $\sqrt{n}$. C is the standard deviation of the sum, $1.2\% \times \sqrt{400}$.

2. **C** is correct: $0.25/(2{,}500 \times 0.02^2) = 0.25$. A is $2\Phi(-2)$, since the miss is $0.02/(0.5/\sqrt{2500}) = 2$ standard deviations: an approximation, not a guarantee. B forgets to square, $\sqrt{0.25}/(\sqrt{2500} \times 0.02) = 0.5$.

3. **B** is correct. With j wins, $W_4/W_0 = 2^{2j-4}$: 16, 4 and 1 at j = 4, 3 and 2. Only $j \geq 3$ exceeds 2.4414, with probability $(1+4)/16 = 5/16$. A counts only four wins. C adds the six paths with $j = 2$, which end at W_0, below the mean.

4. **C** is correct. With $\nu = 1.5 > 1$ the mean exists ($\mu_X = 0$) but the variance is infinite, and the strong law needs only $\mathbb{E}|X| < \infty$. A confuses the two conditions. B needs the finite variance of (17.5).

5. **B** is correct: $5.43/20 = 0.27$ by (17.7). A divides by $\sqrt{20}$, as for a standard error. C assumes that summing keeps the shape. TR23's 20-day sums show 2.29, because its days are not independent (Example 9).

6. The threshold is $0.4150375 \times 400 = 166.015$, so a loss means $j \leq 166$: exactly 0.000394. The normal approximation and its continuity-corrected version give

$$z = -\frac{0.0588915 \times 20}{0.3465736} = -3.3985,$$
$$\Phi(-3.3985) = 0.000339,$$
$$\Phi\Big(\frac{166.5 - 200}{10}\Big) = \Phi(-3.35) = 0.000404.$$

   The uncorrected normal cuts at 166.015, almost half a bar before 166.5, so it omits the normal area 0.000065 between them, about half the bar of $j = 166$. Net of the corrected normal's own overshoot, 0.000010, it falls 0.000055, or 14%, short. At $n = 100$ the cut 41.504 nearly equals 41.5, so the uncorrected value is close to the corrected one. Its remaining 0.7% is mostly the shape's own error, 0.6% for the corrected value.

7. Here $z = 0.055170\% \times \sqrt{1231}/1.2107\% = 1.599$, so $\Pr(W_m < W_0) \approx \Phi(-1.599) = 5.49\%$. The median is $e^{1231 \times 0.00055170} = e^{0.6791} = 1.97$, and the quantiles $e^{0.6791 \mp 0.6988}$ are 0.98 and 3.97, since $1.645 \times 0.012107 \times \sqrt{1231} = 0.6988$. The 5% quantile is below 1 because the loss probability, 5.49%, exceeds 5%: $z = 1.599$ is below 1.645.

8. (a) The event $\{W_m > \mathbb{E}W_m\}$ is $\frac{1}{m}\ln(W_m/W_0) - g > \delta$, which lies inside $\{|\frac{1}{m}\ln(W_m/W_0) - g| \geq \delta\}$. The average has m iid terms with mean g and variance σ_r^2, so (17.2) with $\varepsilon = \delta$ gives the bound. (b) The bound is

$$\frac{0.024288^2}{5{,}849 \times (2.936 \times 10^{-4})^2} = 1.17,$$

   above 1 and empty. The normal approximation gives

$$
\begin{aligned}
z &= \sqrt{5849} \times \frac{2.936\ \text{bp}}{2.4288\%} \\
&= 0.9245, \\
\Phi(-0.9245) &= 0.178,
\end{aligned}
$$

against 0.32 at $f = 1$ (Example 4).

9. (a) Choose the two distinct indices in $m(m-1)/2$ ways, and the two of the four positions that hold Y_i in $\binom{4}{2} = 6$ ways: $6 \times m(m-1)/2 = 3m(m-1)$. (b) Centred, the indicator is $\pm\frac{1}{2}$, so $\mathbb{E}Y^4 = \frac{1}{16} = (\sigma_X^2)^2$: kurtosis 1, excess -2. A binomial count sums m such indicators, so its excess kurtosis is $-2/m$, or -0.1 at $m = 20$, as `stats.binom.stats(20, 0.5, moments="k")` confirms.

10. (a) By (15.9) with $\nu = 1$, $\Pr(|\bar{X}_n| > 1) = 1 - \frac{2}{\pi}\arctan 1 = 1 - \frac{2}{\pi}\cdot\frac{\pi}{4} = \frac{1}{2}$ for every n. For 25 normal draws the average has standard deviation $\frac{1}{5}$, and $2\Phi(-5) = 5.7 \times 10^{-7}$. (b) The first step, (17.1), needs a finite variance, and (14.8) a finite mean and variance. A Cauchy draw has neither.

11. The running share is 0.5163 after 246 returns and 0.5378 after 1,231 (Exhibit 2). The excess kurtosis is 5.43, 3.63, 2.29 and 3.20, against 5.428, 1.086, 0.271 and 0.090 for iid sums (Exhibit 7). It falls far slower than $1/m$, and not at all from 20 to 60 days.

GLOSSARY

Term	中文	Meaning
iid	獨立同分布	Independent and identically distributed: independent draws that share one distribution.
Standard error	標準誤	$\sigma_X/\sqrt{n}$, the standard deviation of an average of n uncorrelated draws, (17.1).
Convergence in probability	依機率收斂	$\Pr(\lvert\bar{X}_n - x\rvert \geq \varepsilon) \to 0$ for every $\varepsilon > 0$.
Weak law of large numbers	弱大數法則	An average of uncorrelated draws with a common mean and a common finite variance converges in probability to the mean, (17.2).
Convergence with probability one	幾乎必然收斂	Convergence along almost every path; the paths where it fails have probability zero together. Also "almost surely".
Strong law of large numbers	強大數法則	For iid draws with a finite mean, the average converges to the mean with probability one.
Gambler's fallacy	賭者謬誤	The belief that independent outcomes correct earlier runs.
Median path	中位數路徑	$W_0 e^{mg}$, approximately the median of wealth, (17.3) and (17.6).
Mean path	期望值路徑	$W_0(1+\mu)^m = \mathbb{E}W_m$, (17.4); ahead of the median path by $e^{m\delta}$.
Convergence in distribution	依分布收斂	Distribution functions converge at every point where the limit's is continuous.
Central limit theorem	中央極限定理	For iid draws with finite positive variance, the standardized sum converges in distribution to $\mathcal{N}(0,1)$, (17.5).
Continuity correction	連續性校正	Cutting a normal approximation of $\Pr(j \leq j_0)$ at $j_0 + \frac{1}{2}$.
Moment generating function	動差母函數	$M_X(t) = \mathbb{E}[e^{tX}]$.
Stable law	穩定分布	A limit of scaled sums of iid draws with infinite variance.

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